Adam Dziedzic

Personal Details

Phone: +1 872 222 8183
Email: adam.dziedzic@sprintml.com
Personal web page: https://adam-dziedzic.com/
Google Scholar: https://scholar.google.com/citations?user=MRk-A4YAAAAJ
LinkedIn: https://www.linkedin.com/in/adziedzic
Twitter: https://twitter.com/adam_dziedzic
GitHub: https://github.com/adam-dziedzic

Research and Work Experience

Current – September 2023
CISPA Helmholtz Center for Information Security, Germany
Tenure Track Faculty Member
My research is focused on secure and trustworthy Machine Learning as a Service (MLaaS). I design robust and reliable machine learning methods for training and inference of ML models while preserving data privacy and model confidentiality.
My Group: SprintML Lab

September 2023 – September 2020
Vector Institute & the University of Toronto, Canada
Postdoctoral Fellow in Computer Science
Group: CleverHans Lab
Advisor: Professor Nicolas Papernot
Research areas: Trustworthy & Collaborative Machine Learning

September 2017 – June 2017
Google (Madison, USA)
PhD Software Engineering Intern at Data Infrastructure and Analysis Team
Mentor: Goetz Graefe

June 2017 – March 2017
Microsoft Research (Redmond, USA)
Research Intern at Data Management, Exploration and Mining (DMX)
Mentors: Vivek Narasayya and Sudipto Das

June 2015 – October 2014
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
Research Intern at Data Intensive Applications and Systems
Advisor: Professor Anastasia Ailamaki

December 2012 – April 2012
CERN (Geneva, Switzerland)
Technical Student at IT Department and CERN Computer Center

August 2013 – June 2013
Barclays Investment Bank (London, The UK)
Analyst at Equities Derivatives Technology

July 2010
Tekten Sp. z o.o. (Warsaw, Poland)
Database designer, Java and PL/SQL software developer

SEPTEMBER 2009 JULY 2009 **Torn** SP. Z O.O. (WARSAW, POLAND)
Java and JavaScript software developer

EDUCATION

AUGUST 2020 JULY 2015 **University of Chicago**, USA
PhD Program in COMPUTER SCIENCE
ADVISOR: Professor Sanjay KRISHNAN
RESEARCH AREAS: Robust Machine & Deep Learning, Data Analysis and Database Systems
GPA: 3.96/4
TEACHING ASSISTANT: Fundamentals of Deep Learning, Introduction to Databases, Databases for Public Policy

MARCH 2013 OCTOBER 2011 **Warsaw University of Technology**, Poland
Master of Science in COMPUTER SCIENCE
MAJOR: Computer Information System Engineering
THESIS: "An analysis and comparison of non-relational (NoSQL) databases with an example of application using CouchDB."
ADVISOR: Professor Piotr GAWRYSIAK
GPA: 4.93/5 (top 5%) THE FINAL GRADE: Excellent

JANUARY 2011 AUGUST 2010 **Technical University of Denmark**
Erasmus Programme
Courses: Logical Systems and Logic Programming, Advanced Databases, Applied Statistics and Statistical Software, Web 2.0 and Mobile Interaction, Java Programming
GPA: 11.71/12

SEPTEMBER 2011 OCTOBER 2007 **Warsaw University of Technology**, Poland
Bachelor of Science in COMPUTER SCIENCE
MAJOR: Computer Information System Engineering
THESIS: "Document management system – application in three-tiered architecture."
ADVISOR: Ph.D. Eng Jarosław DAWIDCZYK
GPA: 4.80/5 (top 5%) THE FINAL GRADE: Excellent

STUDENTS

Current Students:
Hoyong Jeong PhD Student at CISPA (incoming) on *Privacy-Preserving Machine Learning.*
Antoni Kowalczuk PhD Student at CISPA (2023-current) on *Trustworthy Vision Models.*
Bihe Zhao PhD Student at CISPA (2023-current) on *Trustworthy Large Language Models and Dataset Inference.*
Bartłomiej Marek PhD Student at CISPA (2023-current) on *Trustworthy Large Language Models and Privacy Auditing.*
Maitri Shah Research Assistant at CISPA (2025) on *Dataset Provenance for Image Generative Models.*

Ivo Hoese Research Assistant at CISPA (2025) on *Dataset Inference for Image Generative Models.*
Nima DindarSafa Research Assistant at CISPA (2025) on *Dataset Provenance for Large Language Models.*
Stanislaw Cronberg Research Intern at CISPA (2025) on *Trustworthy Large Language Models.*
Filip Szympliński Research Intern at CISPA (2025) on *Dataset Inference for LLMs.*
Filip Manijak Research Intern at CISPA (2025) on *Dataset Inference for Vision.*
Hubert Jastrzębski Research Intern at CISPA (2025) on *Dataset Inference for Audio.*
Aditya Kasliwal Research Intern at CISPA (2025) on *Memorization in Neural Networks.*
Krzysztof Wodnicki Research Intern at CISPA (2025) on *Copyright Identification.*
Piotr Trzaskowski Research Intern at CISPA (2025) on *Copyright Identification.*
Tymoteusz Stępkowski* Research Intern at CISPA (2025) on *Dataset Inference for Vision.*
Jan Kociszewski* Research Intern at CISPA (2025) on *Watermarking for Image Generative Models.*
Krzysztof Rojek* Research Intern at CISPA (2025) on *Watermarking for Image Generative Models.*

***My research interns achieved exceptional success at the International Olympiad in Artificial Intelligence 2025: Krzysztof Rojek (gold medal, 1st place), Tymoteusz Stępkowski (gold medal, 7th place), and Jan Kociszewski (silver medal, 33rd place).**

Past Students:
Jan Dubiński PhD Visiting Research Intern at CISPA (2024), PhD student at Warsaw University of Technology (current) on *Copyright Data Identification in Diffusion Models and Active Defenses Against Model Stealing.*
Olatunji Iyiola Emmanuel Postdoc at CISPA (2023-2025) on *Trustworthy Graph Neural Networks and Privacy of Large Language Models.* Next position: Postdoc at the University of Luxembourg.
Albert Ściseł Research Intern at CISPA (2024) on *Safe Text Generation in Diffusion Models.*
Bartosz Cywiński Research Intern at CISPA (2024) on *Precise Parameter Localization for Textual Generation in Diffusion Models.*
Łukasz Staniszewski Research Intern at CISPA (2024) on *Precise Parameter Localization for Textual Generation in Diffusion Models.*
Shayan Shamsi Research Intern at CISPA (2024) on *Prviacy-Preserving Adaptations of Large Language Models.*
Haonan Duan PhD Student at Vector Institute and the University of Toronto (2022-2023) on *Trustworthy Self-Supervised Learning and Privacy-Preserving Large Language Models.*
Stephan Rabanser PhD Student at Vector Institute and the University of Toronto (2021-2022) on *Trustworthy Machine Learning with a focus on OOD (Out-Of-Distribution) Data.* Next position: Postdoc at Princeton.
Jonas Guan PhD Student at Vector Institute and the University of Toronto (2022) on *Model Stealing and Defenses.*
Yu Shen (Lucy) Lu Member of the CleverHans Lab and an Undergraduate Research Student Assistant at Vector Institute and the University of Toronto (2021) on *Trustworthy Machine Learning with a focus on Model Stealing and Defenses.*
Muhammad Ahmad Kaleem Member of the CleverHans Lab and Research Student at Vector Institute and the University of Toronto (2021-2023) on *Trustworthy Machine Learning with a focus on Model Stealing and Memorization.* Next position: PhD Student at Stanford.

Publications

NeurIPS 2025 — Louis Kerner, Michel Meintz, Bihe Zhao, Franziska Boenisch, **Adam Dziedzic** *BitMark for Infinity: Watermarking Bitwise Autoregressive Image Generative Models* The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.

NeurIPS 2025 — Adarsh Jamadandi, Jing Xu, **Adam Dziedzic**, Franziska Boenisch *Memorization in Graph Neural Networks* The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.

NeurIPS 2025 — Jamie Hayes, **Adam Dziedzic**, A. Feder Cooper, Christopher A. Choquette-Choo, Franziska Boenisch, Georgios Kaissis, Igor Shilov, Ilia Shumailov, Katherine Lee, Matthew Jagielski, Matthieu Meeus, Meenatchi Sundaram Muthu Selva Annamalai, Niloofar Mireshghallah, Yves-Alexandre de Montjoye, Milad Nasr *Strong Membership Inference Attacks on Massive Datasets and (Moderately) Large Language Models* The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.

ICML 2025 — Antoni Kowalczuk, Jan Dubiński, Franziska Boenisch, **Adam Dziedzic** *Privacy Attacks on Image AutoRegressive Models* The Forty-Second International Conference on Machine Learning (ICML), 2025.

ICML 2025 — Bihe Zhao, Pratyush Maini, Franziska Boenisch, **Adam Dziedzic** *Unlocking Post-hoc Dataset Inference with Synthetic Data* The Forty-Second International Conference on Machine Learning (ICML), 2025.

ICML 2025 — Xun Wang, Jing Xu, Franziska Boenisch, Michael Backes, Christopher A. Choquette-Choo, **Adam Dziedzic** *Efficient and Privacy-Preserving Soft Prompt Transfer for LLMs* The Forty-Second International Conference on Machine Learning (ICML), 2025.

CCS 2025 — Olive Franzese, Congyu Fang, Radhika Ghard, Xiao Wang, Somesh Jha, Nicolas Papernot, **Adam Dziedzic** *Secure Noise Sampling for Differentially Private Collaborative Learning* The 32nd ACM Conference on Computer and Communications Security (CCS), 2025.

CVPR 2025 — Jan Dubiński, Antoni Kowalczuk, Franziska Boenisch, **Adam Dziedzic** *CDI: Copyrighted Data Identification in Diffusion Models* The IEEE CVF Computer Vision and Pattern Recognition Conference (CVPR), 2025.

ICLR 2025 — Łukasz Staniszewski, Bartosz Cywiński, Franziska Boenisch, Kamil Deja, **Adam Dziedzic** *Precise Parameter Localization for Textual Generation in Diffusion Models* The Thirteenth International Conference on Learning Representations, 2025.

ICLR 2025 — Wenhao Wang, **Adam Dziedzic**, Grace C. Kim, Michael Backes, Franziska Boenisch *Captured by Captions: On Memorization and its Mitigation in CLIP Models* The Thirteenth International Conference on Learning Representations, 2025.

ICLR 2025	Shahrzad Kiani, Nupur Kulkarni, **Adam Dziedzic**, Stark Draper, Franziska Boenisch *Differentially Private Federated Learning with Time-Adaptive Privacy Spending* The Thirteenth International Conference on Learning Representations, 2025.
AAAI 2025	Dariush Wahdany, Matthew Jagielski, **Adam Dziedzic**, Franziska Boenisch *Beyond the Mean: Differentially Private Prototypes for Private Transfer Learning* The Association for the Advancement of Artificial Intelligence, 2025.
TMLR 2025	Stephan Rabanser, Anvith Thudi, Kimia Hamidieh, **Adam Dziedzic**, Israfil Bahceci, Akram Bin Sediq, Hamza Sokun, Nicolas Papernot *Selective Prediction via Training Dynamics* Transactions on Machine Learning Research, 2025.
NeurIPS 2024	Vincent Hanke, Tom Blanchard, Franziska Boenisch, Iyiola Emmanuel Olatunji, Michael Backes, **Adam Dziedzic** *Open LLMs are Necessary for Current Private Adaptations and Outperform their Closed Alternatives* The Thirty-Eighth Annual Conference on Neural Information Processing Systems, 2024.
NeurIPS 2024	Pratyush Maini, Hengrui Jia, Nicolas Papernot, **Adam Dziedzic** *LLM Dataset Inference: Did you train on my dataset?* The Thirty-Eighth Annual Conference on Neural Information Processing Systems, 2024.
NeurIPS 2024	Wenhao Wang, **Adam Dziedzic**, Michael Backes, Franziska Boenisch *Localizing Memorization in SSL Vision Encoders* The Thirty-Eighth Annual Conference on Neural Information Processing Systems, 2024.
NeurIPS 2024	Dominik Hintersdorf, Lukas Struppek, Kristian Kersting, **Adam Dziedzic**, Franziska Boenisch *Finding NeMo: Localizing Neurons Responsible For Memorization in Diffusion Models* The Thirty-Eighth Annual Conference on Neural Information Processing Systems, 2024.
eBioMedicine 2024	Congyu Fang, **Adam Dziedzic**, Lin Zhang, Laura Oliva, Amol Verma, Fahad Razak, Nicolas Papernot, Bo Wang *Decentralised, Collaborative, and Privacy-preserving Machine Learning for Multi-Hospital Data* EBioMedicine 101, 2024.
ICLR 2024	Wenhao Wang, Muhammad Ahmad Kaleem, **Adam Dziedzic**, Michael Backes, Nicolas Papernot, Franziska Boenisch *Memorization in Self-Supervised Learning Improves Downstream Generalization* The Twelfth International Conference on Learning Representations, 2024.
NeurIPS 2023	Jan Dubiński, Stanisław Pawlak, Franziska Boenisch, Tomasz Trzcinski, **Adam Dziedzic** *Bucks for Buckets (B4B): Active Defenses Against Stealing Encoders* The Thirty-Seventh Annual Conference on Neural Information Processing Systems, 2023.

NeurIPS 2023	Nicholas Franzese, **Adam Dziedzic**, Christopher A. Choquette-Choo, Mark R. Thomas, Muhammad Ahmad Kaleem, Stephan Rabanser, Congyu Fang, Somesh Jha, Nicolas Papernot, Xiao Wang *Robust and Actively Secure Serverless Collaborative Learning* The Thirty-Seventh Annual Conference on Neural Information Processing Systems, 2023.
NeurIPS 2023	Haonan Duan, **Adam Dziedzic**, Nicolas Papernot, Franziska Boenisch *Flocks of Stochastic Parrots: Differentially Private Prompt Learning for Large Language Models* The Thirty-Seventh Annual Conference on Neural Information Processing Systems, 2023.
NeurIPS 2023	Franziska Boenisch, Christopher Mühl, **Adam Dziedzic**, Roy Rinberg, Nicolas Papernot *Have it your way: Individualized Privacy Assignment for DP-SGD* The Thirty-Seventh Annual Conference on Neural Information Processing Systems, 2023.
PETs 2023	**Adam Dziedzic**, Christopher A. Choquette-Choo, Natalie Dullerud, Vinith Suriyakumar, Ali Shahin Shamsabadi, Muhammad Ahmad Kaleem, Somesh Jha, Nicolas Papernot, Xiao Wang *Private Multi-Winner Voting for Machine Learning* The annual Privacy Enhancing Technologies Symposium (PETS), 2023.
PETs 2023	Franziska Boenisch, Christopher Mühl, Roy Rinberg, Jannis Ihrig, **Adam Dziedzic** *Individualized PATE: Differentially Private Machine Learning with Individual Privacy Guarantees* The annual Privacy Enhancing Technologies Symposium (PETS), 2023.
EuroS&P 2023	Franziska Boenisch, **Adam Dziedzic**, Roei Schuster, Ali Shahin Shamsabadi, Ilia Shumailov, Nicolas Papernot *When the Curious Abandon Honesty: Federated Learning Is Not Private* IEEE European Symposium on Security and Privacy, 2023.
EuroS&P 2023	Franziska Boenisch, **Adam Dziedzic**, Roei Schuster, Ali Shahin Shamsabadi, Ilia Shumailov, Nicolas Papernot *Is Federated Learning a Practical PET Yet?* IEEE European Symposium on Security and Privacy, 2023.
ICLR Workshop 2023	**Adam Dziedzic**, Franziska Boenisch, Mingjian Jiang, Haonan Duan, Nicolas Papernot *Sentence Embedding Encoders are Easy to Steal but Hard to Defend* ICLR (The Eleventh International Conference on Learning Representations) Workshop on Trustworthy ML, 2023.
ICLR 2022	**Adam Dziedzic**, Muhammad Ahmad Kaleem, Yu Shen Lu, Nicolas Papernot *Increasing the Cost of Model Extraction with Calibrated Proof of Work* **SPOTLIGHT** (top 5% of accepted papers) The Tenth International Conference on Learning Representations, 2022.
NeurIPS 2022	**Adam Dziedzic**, Haonan Duan, Muhammad Ahmad Kaleem, Nikita Dhawan, Jonas Guan, Yannis Cattan, Franziska Boenisch, Nicolas Papernot *Dataset Inference for Self-Supervised Models* The Thirty-Sixth Annual Conference on Neural Information Processing Systems, 2022.

ICML 2022	**Adam Dziedzic**, Nikita Dhawan, Muhammad Ahmad Kaleem, Jonas Guan, Nicolas Papernot *On the Difficulty of Defending Self-Supervised Learning against Model Extraction* The Thirty-Ninth International Conference on Machine Learning, 2022.
ICLR 2021	Christopher A. Choquette-Choo, Natalie Dullerud, **Adam Dziedzic**, Yunxiang Zhang, Somesh Jha, Nicolas Papernot, Xiao Wang *CaPC Learning: Confidential and Private Collaborative Learning* The Ninth International Conference on Learning Representations, 2021.
ACL 2020	Dan Hendrycks, Xiaoyuan Liu, Eric Wallace, **Adam Dziedzic**, Rishabh Krishnan, Dawn Song *Pretrained Transformers Improve Out-of-Distribution Robustness* The 58th Annual Meeting of the Association for Computational Linguistics, 2020.
JOR 2020	Arnold Wong, Garrett Harada, Remy Lee, Sapan D. Gandhi, **Adam Dziedzic**, Alejandro Espinoza-Orias, Mohamad Parnianpour, Philip Louie, Bryce Basques, Howard S. An, Dino Samartzis *Preoperative paraspinal neck muscle characteristics predict early-onset adjacent segment degeneration in anterior cervical fusion patients: a machine-learning modeling analysis* Journal of Orthopaedic Research® 39.8 (2021): 1732-1744.
OJVT 2020	**Adam Dziedzic**, Vanlin Sathya, Monisha Ghosh, Sanjay Krishnan *Machine Learning for Fair Spectrum Sharing in Dense LTE Wi-Fi Coexistence* IEEE Open Journal of Vehicular Technology, 2020.
ICNC 2020	Vanlin Sathya, **Adam Dziedzic**, Monisha Ghosh, Sanjay Krishnan *Machine learning-based detection of multiple Wi-Fi BSSs for LTE-U CSAT* International Conference on Computing, Networking and Communication (ICNC), 2020.
Ph.D. 2020	**Adam Dziedzic** *Input and Model Compression for Adaptive and Robust Neural Networks* (Ph.D. Thesis)
ICML 2019	**Adam Dziedzic**, Ioannis Paparrizos, Sanjay Krishnan, Aaron J. Elmore, Michael Franklin *Band-limited Training and Inference for Convolutional Neural Networks* (paper) code: https://github.com/adam-dziedzic/bandlimited-cnns The Thirty-Sixth International Conference on Machine Learning, 2019.
SIGOPS 2019	Sanjay Krishnan, Aaron J. Elmore, Michael Franklin, Ioannis Paparrizos, Zechao Shang, **Adam Dziedzic**, Rui Liu *Artificial Intelligence in Resource-Constrained and Shared Environments* ACM Special Interest Group in Operating Systems (SIGOPS), 2019.
CIDR 2019	Sanjay Krishnan, **Adam Dziedzic**, Aaron J. Elmore *DeepLens: Towards a Visual Data Management System* Conference on Innovative Data Systems Research (CIDR), 2019.
SIGMOD 2018	**Adam Dziedzic**, Jingjing Wang, Sudipto Das, Bolin Ding, Vivek R. Narasayya, Manoj Syamala *Columnstore and B+ tree – Are Hybrid Physical Designs Important?* The 2018 International Conference on Management of Data, 2018.

UChicago 2017	**Adam Dziedzic** *Data Loading, Transformation, and Migration for Database Management Systems* (Master's thesis)
CIDR 2017	Tim Mattson, Vijay Gadepally, Zuohao She, **Adam Dziedzic**, Jeff Parkhurst *Demonstrating the BigDAWG Polystore System for Ocean Metagenomic Analysis* Conference on Innovative Data Systems Research (CIDR), 2017.
VLDB ADMS 2016	**Adam Dziedzic**, Manos Karpathiotakis, Ioannis Alagiannis, Raja Appuswamy, Anastasia Ailamaki *DBMS Data Loading: An Analysis on Modern Hardware* Data Management on New Hardware, 2016.
HPEC 2016	**Adam Dziedzic**, Aaron J. Elmore, Michael Stonebraker *Data Transformation and Migration in Polystores* (paper) code: https://github.com/bigdawg-istc/bigdawg IEEE High Performance Extreme Computing Conference, 2016.
HPEC 2016	John Meehan, Stan Zdonik, Shaobo Tian, Yulong Tian, Nesime Tatbul, **Adam Dziedzic** and Aaron J. Elmore *Integrating Real-Time and Batch Processing in a Polystore* IEEE High Performance Extreme Computing Conference, 2016.
IEEE VIS DSIA 2015	**Adam Dziedzic**, Jennie Duggan, Aaron J. Elmore, Vijay Gadepally, Michael Stonebraker *BigDAWG: a Polystore for Diverse Interactive Applications* The IEEE Visualization Conference (VIS), 2015.
SPIE 2014	**Adam Dziedzic**, Jan Mulawka. *Analysis and Comparison of databases with an introduction to consistent references in big data storage systems* Photonics Applications in Astronomy, Communications, Industry, and High-Energy Physics Experiments, 2014.

Preprints

ArXiv 2022	Stephan Rabanser, Anvith Thudi, Kimia Hamidieh, **Adam Dziedzic**, Nicolas Papernot *Selective Classification Via Neural Network Training Dynamics*
ArXiv 2022	**Adam Dziedzic**, Stephan Rabanser, Mohammad Yaghini, Armin Ale, Murat A. Erdogdu, Nicolas Papernot *p-DkNN: Out-of-Distribution Detection Through Statistical Testing of Deep Representations*
ArXiv 2021	Adelin Travers, Lorna Licollari, Guanghan Wang, Varun Chandrasekaran, **Adam Dziedzic**, David Lie, Nicolas Papernot *On the Exploitability of Audio Machine Learning Pipelines to Surreptitious Adversarial Examples*
Intel 2021	Ahmad-Reza Sadeghi, Ferdinand Brasser, Markus Miettinen, Thien Duc Nguyen, Thomas Given-Wilson, Axel Legay, Murali Annaaram, Salman Avestimeh, Alexandra Dmitrienko, Farinaz Koushanfar, Buse Gul Atli, Florian Kerschbaum, Lachlan J. Gunn, N. Asokan, Matthias Schunter, Rosario Cammarota, **Adam Dziedzic**, Nicolas Papernot, Virginia Smith, Reza Shokri *Private AI Collaborative Research Institute: Vision, Challenges, and Opportunities*
ArXiv 2020	**Adam Dziedzic**, Sanjay Krishnan *Empirical Evaluation of Perturbation-based Defenses*

Teaching

Trustworthy Machine Learning 2025 (Lecture)	TML: Advanced Lecture on Trustworthy Machine Learning at the Saarland University and CISPA given at the summer semester 2025. This course explores the different aspects of trustworthy machine learning, including Privacy, Collaborative Learning, Model Confidentiality, Robustness, Fairness and Bias, Explainability, Security, and Governance.
Trustworthy Machine Learning 2024 (Seminar)	TrustML: Seminar on Trustworthy Machine Learning at the Saarland University and CISPA given in the winter semester 2024/2025. The main focus of the seminar is on the security, privacy, confidentiality, and robustness of machine learning models.
Trustworthy Machine Learning 2024 (Lecture)	TML: Advanced Lecture on Trustworthy Machine Learning at the Saarland University and CISPA given at the summer semester 2024. This course explores the different aspects of trustworthy machine learning, including Privacy, Collaborative Learning, Model Confidentiality, Robustness, Fairness and Bias, Explainability, Security, and Governance.
Trustworthy Machine Learning 2023 (Seminar)	TrustML: Seminar on Trustworthy Machine Learning at the Saarland University and CISPA given in the winter semester 2023/2024. The main focus of the seminar is on the security, privacy, confidentiality, and robustness of machine learning models.
Deep Learning	*TTIC-31230: Teaching assistant for the course on Fundamentals of Deep Learning taught by Prof. David McAllester* (Winter 2020)

Database Systems	*CS23500/33550: Teaching assistant for the course on Database Systems taught by Prof. Aaron J. Elmore* (Autumn 2015, Spring 2016, Winter 2017, Winter 2018, Spring 2018)
Bioinformatics Algorithms	*MBI: Teaching assistant for the course on Methods in Bioinformatics taught by Prof. Robert M. Nowak* (Spring 2014)

Selected Talks

2025	**Baltic Energy and Cyber Security Forum** in the Embassy of the Republic of Poland in Berlin (June 17th, 2025) on *"Model Stealing and Defenses"*.
2025	**Google TechTalk** (May 28th, 2025) on *"Private Adaptations of Large Language Models"*.
2024	**Tea talk at MILA** (Montreal Institute for Learning Algorithms) on *"Private Adaptations of Large Language Models"* (November 29th, 2024).
2024	Gave an invited talk at **Intel** on *"Private Adaptations of Open LLMs Outperform their Closed Alternatives"*.
2024	Gave a presentation at the ML in PL conference (Machine Learning Conference in Poland) on Private Adaptations of Open LLMs Outperform their Closed Alternatives (November 8th).
2024	Talk at Google for the ML Red Team Knowledge Sharing on **Private Adaptations of Large Language Models** (October 28th).
2024	Talk at NUS (National University of Singapore) on Private Prompt Learning for Large Language Models (July 1st).
2024	Talk at A*STAR Institute for Infocomm Research on **Private Prompt Learning for Large Language Models**. I was hosted by Dr. Dinil Mon Divakaran (July 1st).
2024	Gave a talk on *Private Prompt Learning for Large Language Models* at the Machine Learning Security Seminar Series (May 29th).
2024	Gave a seminar on *Private Prompt Learning for Large Language Models* at the University of Waterloo (April 12th).
2023	Talk about our paper on Flocks of Stochastic Parrots: Differentially Private Prompt Learning for Large Language Models at IDEAS NCBR (December).
2023	Talk on model stealing and defenses and speaker at the panel discussion at the NeurIPS workshop on Backdoors in Deep Learning: The Good, the Bad, and the Ugly.
2023	Presentation of our paper on Flocks of Stochastic Parrots: Differentially Private Prompt Learning for Large Language Models at MLinPL 2023 (October 27th).
2023	Presentation of our paper on Private Multi-Winner Voting for Machine Learning at PETS 2023 (July 12th).
2023	Talk on "Is this model mine? On stealing and defending machine learning models."for ML capstone class ECE697 from the University of Wisconsin-Madison (June 20th).
2023	Podcast Interview: On model stealing and defenses hosted by Matt Faltyn. https://traincheck.buzzsprout.com/
2023	Is this model mine? On stealing and defending machine learning models. CISPA, Germany.

2022 Is this Encoder Mine? On Stealing and Defending Self-Supervised Encoders
AI Safety Unconference NeurIPS 2022

2022 Stealing and Defending Self-Supervised Models. Invited talk on Dataset Inference for Self-Supervised Models at **ML Collective** (a nonprofit research organization) during their reading group "Deep Learning: Classics and Trends"which runs weekly, is fully virtual and is open to the public. They have 3000+ email subscribers and on average 100 weekly attendees.

2022 **Managing AI Risk - Cybersecurity & Data Risk Workstream at Vector Institute:** Is this Encoder Mine? On Stealing and Defending Self-Supervised Encoders

2022 Is this model mine? On stealing and defending machine learning models
University of Michigan at Ann Arbor

2022 Collaborative Machine Learning.
Vector Talk Series

2021 Confidential and Private Collaborative Learning.
Scotia Bank - Research Frontier Talk Series

2021 CaPC Learning: Confidential and Private Collaborative Learning.
Vector School: AI Model Governance

2021 CaPC Learning: Confidential and Private Collaborative Learning.
Invited Speaker for the Third Workshop on Privacy in Natural Language Processing.

2021 CaPC Learning: Confidential and Private Collaborative Learning.
The MLFL series, hosted by the Center for Data Science, UMass Amherst.

2021 CaPC Learning: Confidential and Private Collaborative Learning.
Flow Seminar

2021 CaPC Learning: Confidential and Private Collaborative Learning.
Intel Labs

2020 CaPC Learning: Confidential and Private Collaborative Learning.
Vector Institute

2019 Talk at the ICML 2019 conference on **Band-limited Training and Inference for Convolutional Neural Networks**.

2018 Columnstore and B+ tree – are hybrid physical designs important?
University of California, Berkeley

2018 Columnstore and B+ tree – are hybrid physical designs important?
Imperial College London

2018 Columnstore and B+ tree – are hybrid physical designs important?
Oracle

2018 Columnstore and B+ tree – are hybrid physical designs important?
Microsoft Research

2018 Columnstore and B+ tree – are hybrid physical designs important?
MemSQL

2018 Talk at the SIGMOD 2018 Conference on **Columnstore and B+ tree – are hybrid physical designs important?**

2017 Talk at the Microsoft Research in Redmond on **Columnstore and B+ tree – are hybrid physical designs important?**

2016 Talk at HPEC (IEEE High Performance Extreme Computing) on Data Transformation and Migration in Polystores.

2015	Talk at IEEE Viz Data Systems for Interactive Analysis in Chicago on BigDAWG: a Polystore for Diverse Interactive Applications.

Awards and Grants

2025	Cybersecurity Grant from OpenAI.
2025	Unrestricted Gift from G-Research.
2024	DFG Weave-OPUS-LAP Grant (funding for 3 years and 1 PhD student), project title: "Protecting Creativity: Towards Safe Generative Models".
2024	The Best Poster Award at the MLinPL (Machine Learning in Poland) Conference for our work on *Ready, Aim, Edit! Precise Parameter Localization for Text Editing with Diffusion Models.*.
2023	The Best Poster Award at the MLinPL (Machine Learning in Poland) Conference for our work on *Bucks for Buckets (B4B): Active Defenses Against Stealing Encoders*.
2022	Spotlight Paper on a new defense against model extraction at International Conference on Learning Representations (ICLR).
2022	Highlighted Reviewer at the International Conference on Learning Representations (ICLR).
2019	Travel Award at International Conference on Machine Learning (ICML).
2018	Travel Award at SIGMOD (Special Interest Group on Management of Data).
2011-2012	The academic scholarship of the Rector of the Warsaw University of Technology for my achievements during the Master's program.
2007-2011	The academic scholarship of the Rector of the Warsaw University of Technology for the best faculty students (granted on a yearly basis and based on GPA).

Service and Volunteering

Workshops	Co-organized at ICML 2025: (1) The Impact of Memorization on Trustworthy Foundation Models (Understanding unintended memorization is essential to building trustworthy foundation models.) (2) Data in Generative Models (Building reliable and trustworthy Generative AI with high-quality and responsibly managed training data.)
Hackathons	Co-organized two hackathons in Warsaw, Poland in March 2024 and in Cracow, Poland in March 2025. There were 200 participants at each event and the winning team was offered internships at the SprintML Lab. The tasks at the hackathons were based on our research on Trustworthy Machine Learning.
WiML	I serve as a Mentor during the Women in Machine Learning sessions at ICML, NeurIPS, and ICLR conferences.
SaTML	Served as a session chair at the SaTML 2023 and 2024 conferences for the session on Collaborative learning. Reviewer at the conference: 2023, 2024, 2025.
Vector	Served on the Research Adjudication Committee for the Vector Scholarship in Artificial Intelligence: 2022.
S&P	Program Committee Member: 2025.
USENIX	Program Committee Member: 2023.
CCS	Program Committee Member: 2023 in the Machine Learning and Security Track, 2024 in the Machine Learning and Security Track.
ICLR	Area Chair (2025). Reviewer at the International Conference on Learning Representations: 2019, 2020, 2021, **2022 highlighted reviewer, top 5%**, 2023, 2024.

ICML	Area Chair (2025). Reviewer at the International Conference on Machine Learning: 2021, 2022, 2023, 2024.
NeurIPS	Area Chair (2025). Reviewer at the conference on Neural Information Processing Systems: 2021, 2022, 2023, 2024.

References

Nicolas Papernot	Assistant Professor at the University of Toronto and the Vector Institute email: nicolas.papernot@utoronto.ca
Sanjay Krishnan	Assistant Professor at the University of Chicago email: skr@uchicago.edu
Tomasz Trzciński	Full Professor at Warsaw University of Technology and a leader of the Computer Vision Group at IDEAS NCBR, a publicly-funded Polish Center for Artificial Intelligence email: tomasz.trzcinski@pw.edu.pl
Somesh Jha	Lubar Professor at the University of Wisconsin, Madison email: jha@cs.wisc.edu
Xiao Wang	Assistant Professor at Northwestern University email: wangxiao@cs.northwestern.edu
Vivek Narasayya	Principal Researcher at Microsoft Research, Redmond email: viveknar@microsoft.com
Michael Franklin	Liew Family Chairman of Computer Science at the University of Chicago email: mjfranklin@uchicago.edu